Geographic Immobility in the United States: Assessing the Prevalence and Characteristics of Those Who Never Migrate Across State Lines Using Linked Federal Tax Microdata

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Abstract

This paper explores the prevalence and characteristics of those who never migrate at the state scale in the U.S. Studying people who never migrate requires regular and frequent observation of their residential location for a lifetime, or at least for many years. A novel U.S. population-sized longitudinal dataset that links individual level Internal Revenue Service (IRS) and Social Security Administration (SSA) administrative records supplies this information annually, along with information on income and socio-demographic characteristics. We use these administrative microdata to follow a cohort aged between 15 and 50 in 2001 from 2001 to 2016, differentiating those who lived in the same state every year during this period (i.e., never made an interstate move) from those who lived in more than one state (i.e., made at least one interstate move). We find those who never made an interstate move comprised 75 percent of the total population of this age cohort. This percentage varies by year of age but never falls below 62 percent even for those who were teenagers or young adults in 2001. There are also variations in these percentages by sex, race, nativity, and income, with the latter having the largest effects. We also find substantial variation in these percentages across states. Our findings suggest a need for more research on geographically immobile populations in U.S.

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INTRODUCTION

The migration rate, the fraction of the population who move in a time interval, typically one or five years, is the standard measure of migration activity in a population (Long 1991; Bell et.al, 2015a). Variations in this rate over time and across space are the foundation on which comparisons of the relative intensity of migration in populations are made (Bell et. al, 2015b). These rates are not the only way to gauge migration activity. An alternative is to assess the prevalence of migration *experience* in the population, defined as the fraction of the population that has *ever* migrated (i.e. migrated at least once). Importantly, this fraction renders visible its remainder, the fraction that has *never* migrated. These "never movers" in the U.S. are the subject of this paper.

Those who never migrate are seldom the subject of migration research (Hjälm 2014; Schewel, 2019; Erickson *et al.*, 2018). Arguably, this has been particularly true in the U.S. where a high rate of geographic mobility has been a persistent thread in descriptions of American life since the early nineteenth century (Tocqueville, 2018; Fischer 2002; Hall and Ruggles, 2004; Ferrie 2005). The well-documented decline in U.S. migration rates since the 1980s weakened the power of this narrative, prompting descriptions of contemporary Americans as increasingly rooted (Cooke, 2011; Foster, 2017; Peng and Butts 2023). Our goal is to build on the recent attention to geographic rootedness in American life by identifying the prevalence and characteristics of those who never migrate. The extent to which people never move is perhaps a more fundamental challenge than declining rates of migration to the notion that the experience of migration is a defining feature of U.S. society.

It is important to define at the outset who never movers are and who they are not. Never movers are not those who do not move in typical migration rate time intervals. These non-migrants, often called stayers, are simply those who do not move in the same one- or five-year period as those who migrate during this period. Never movers are a subset of this stayer population, but this population also includes those who moved before this time interval and those who will do so later.

Never movers are also not the same as stayers derived from lifetime migration studies. These are people recorded at some point later in their life as living in their place or area of birth. These studies typically use two locations, place/area of birth and current residential location, to determine whether someone has moved or stayed, with little or no information on where they lived in the intervening period (e.g. Eldridge and Thomas, 1964; O'Brien and Larmon, 1968;

Newbold, 2001 Bell and Muhidin, 2009). As above, never movers are a subset of this lifetime migration-defined stayer population, but this population also includes those who have migrated at least twice, with the final move a return to their place of birth.

Distinguishing never movers is more burdensome in data requirements than identifying either of the forms of stayers just described because it requires a longitudinal record of all the places people have lived, or at least a multiyear annual record of these places. Ideally this record would be for a lifetime but at a minimum should be available for a long enough period to yield a meaningful assessment of never moving. Population registries can supply this information but in countries without them it is only available through surveys that are hard and expensive to collect because of follow-up and attrition issues. As a result, in the U.S. at least, suitable longitudinal data tend only to be available in relatively small sample surveys ($N < 20{,}000$), such as the Panel Study of Income Dynamics, that constrain the possibilities for detailed subgroup and geographic analysis of migration experience, or its absence.

These data issues are likely part of the reason why there has been little attention to ever migrating or never migrating as alternate measures of geographic (im)mobility in the U.S. A novel population-sized longitudinal administrative dataset built from linked individual level Internal Revenue Service (IRS) and Social Security Administration (SSA) administrative records provides an opportunity to calculate these measures. These data provide annual records on the residential location of all those who file taxes and their dependents; they also include this locational information for those who do not file but have income reported to the IRS through information returns. Information from these tax filings and information returns and the SSA also provide us with information on the demographic and economic characteristics of these people.

We use these data to investigate never migrating in the U.S., focusing on those between 15 and 50 years old in 2001 (30-65 in 2016) who never moved across state lines between 2001 and 2016. We focus on this age range because it includes the young adult ages when peak migration intensity occurs, tailing off in later work years. Specifically, we seek answers to the following questions: What is the prevalence of never migrating across state lines for people who are in the age range when migration intensity is highest? How does this prevalence vary by age, sex, nativity, race, and income? And how does it vary across states? Before presenting our empirical analysis of these questions we first review and evaluate literatures on migration and immobility and then describe the data and how we use it to construct our locational histories and measure of never migrating. Following the analysis, we conclude by summarizing key findings and

suggesting directions for future research on immobility in the U.S.

BACKGROUND AND FRAMING

Migration and immobility in the U.S.

Toqueville's observations on the restlessness of Americans during his travels in the U.S. in the 1830s is commonly the starting point for claims that high rates of migration are one manifestation of American exceptionalism (Tocqueville, 2018, Ferrie, 2005). Similar depictions continued well into the second half of the twentieth century (e.g., Packard, 1972) despite migration rates in this era being lower than in the previous century (Fischer, 2002; Hall and Ruggles, 2004). Thus, while Americans had become more rooted in the second half of the twentieth century the mythology of their restlessness and its consequences, desirable or not, lived on. Only in the last two decades, after migration rates had declined by 50 percent from their post Second World War peak (Cooke, 2013; Kaplan and Schulhofer-Wohl, 2017), has this mythology of American migration exceptionalism become hard to sustain.

Research on the migration decline in the U.S. - and in other countries where falling migration rates have also been observed - identifies several reasons for why people are moving less now than a few decades ago. These include reductions in the frequency of job changing, increases in economic precarity, the substitutability of other forms of temporary mobility for permanent moves, and a general shift in behavior towards rootedness (Cooke, 2011; Molloy et.al, 2011; Foster, 2017; Molloy *et al.*, 2017; Cooke and Shuttleworth 2018; Preece 2018; Kalemba *et al.*, 2020; White and Haan 2021; Kalemba *et al.*, 2022; Coulter 2023). In the U.S., the migration decline and a growing regional divide between those areas prospering post Great Recession and those not has prompted media and scholarly narratives of populations “stuck” in place or “left-behind” who are unable or unwilling to move to access better economic opportunities (Johnson, Curtis, and Egan-Robertson 2017; Adamy and Overberg 2017; Feintzeig and Weber 2018; Wuthnow, 2019, Partridge et al., 2020; Pike et al., 2023; Eisenberg, 2024). Our goal here is not to investigate this decline further or adjudicate the debate about possible constraints on U.S. internal migration. Rather, the light these exchanges have shone on the issue of those who are rooted or stuck, who never migrate, is a motivation for our project.

Most assessments of the U.S. migration decline are based on two cross-sectional surveys: the Current Population Survey (CPS) and the American Community Survey (ACS). While these data provide annual measures of migration rates over time (since 1948 for the CPS and 2005 for the

ACS) they cannot gauge who *never* migrates or who stays in place for long periods of time, possibly lifetimes (i.e. the immobile who most directly equate to being stuck or rooted). These data and historical U.S. decennial census records also provide information on birthplace, making it possible to measure changes in the percentage of those living in and outside their state of birth since 1850 (Hall and Ruggles, 2004). As already noted, there will be return migrants living in their state of birth meaning these data cannot reveal the extent of never migrating from each state. The other major source of U.S. migration data, publicly available IRS migration data, have similar limitations.

Existing longitudinal data sources, such as the Panel Study of Income Dynamics, Survey of Income and Program Participation, the National Longitudinal Study and the Health and Retirement Survey, can provide insight on those who never migrate (Sharkey, 2015; Bernard *et al.*, 2017). As rich as these data are, however, their small sample sizes limit the availability, and when available the statistical precision, of these events for population subgroups differentiated by socio-demographic characteristics and places (Kaplan and Schulhofer-Wohl 2017). The IRS administrative microdata we introduce here, linked to other records that provide information on age, race, sex, nativity, and citizenship status, do not have this small sample problem.

Despite their limitations, findings from small sample longitudinal and other sources are suggestive of experiences that counter the narrative of migration as a central feature of American life (Erickson *et al.*, 2018). As such, they provide hints about what to expect from the novel administrative data we use. Sharkey (2015) used the Panel Study of Income Dynamics to study the location of Black and White families across four generations. Most striking is his finding of a high rate of remaining in the same county or state for both groups: at least half of Black families remained in the same county across generations; just under half of White families did the same. At the state scale, these proportions rise to 75% or more of both groups of families remaining in the same state.

Part of the popular narrative of American migratory experiences is that people not only move frequently but also move far, with movers residing at great distances from their relatives. We know most moves in the U.S. are actually short-distance relocations, less than 50 km or within counties (Long *et al.* 1988). Given this, it is not surprising that data from the National Survey of Families and Household reveals that the median distance between married heterosexual couples and their parents is 25 miles for the male partner from his parents and 20 miles for the female partner from hers (Compton and Pollak 2015). These distances are short enough that many will reside within the same county as their parents, and a large fraction within the same

state. If migrating is defined as leaving these jurisdictions, or as a move that relocates someone far enough away to disrupt the frequency of pre-move contact with family and other network interaction (e.g. > 30 miles/50km), then these parent-child relative locations are consistent with a definition of never migrating.

Immobility and migration studies

There is a vast literature on migration documenting who movers are and why they move. Except for those forced to move because of violence or environmental catastrophe, migrants tend to be young adults with above-average education in the communities in which they originate (e.g., Thomas 1938; Sjaastad 1962; Long 1973; Rogers et al. 1978; Greenwood 1975). Those with connections to people in other places are also more likely to migrate (e.g., Massey et al. 1987; Palloni et al. 2001; Haug 2008). Migration theories explain why the benefits of migration outweigh the economic and social costs of moving for these population subgroups (e.g., Sjaastad 1962, Greenwood 1985; Massey et al., 1993).

In framings of migration that start from these orientations, those who are not in these selective subgroups, such as older adults, are not expected to move. What of the young who are expected to move but do not? Research shows that economic conditions where people live can affect the propensity to leave, explaining why those even in the most mobile subgroups might be apt to stay put in some places but less so in others (Greenwood 1975, 1985). It also shows that even when those conditions are unfavorable some people do not move, possibly because they are constrained from actualizing a migration by lack of resources to make the move or by poor information about alternatives elsewhere (Clark 1983). From this perspective, non-movers are stuck and would migrate if they could, i.e., if the barriers to mobility were eliminated (Adamy and Overberg 2017; Schleicher 2017). There is an alternative view of not migrating that elevates the significance of the social and psychological benefits of proximity to family and friends and connections to community (Imbroscio 2004). Those for whom place-based benefits are most important do not migrate because they are rooted to a place rather than stuck (Cooke 2011; Huang and Butts 2023)

Those who choose to stay put have never been a central focus of migration research, an absence that prompted Schewel (2019) to argue that migration research has a mobility bias, a conceptual and empirical orientation to what motivates people to migrate rather than a broader interest in their aspirations and capabilities to either migrate *or* stay (see also de Haas, 2021). The bias originates in part from the idea that migration is an action that represents a choice to

depart from the norm or the default status of being immobile (Schewel 2019). Staying where one is, being immobile, can be a deliberate or positive choice too, or a freedom as de Haas (2021) puts it (Barcus and Brunn 2009, 2010; Hjälm 2014; Schewel 2019; Mata-Codesal 2018; Stockdale and Haartsen, 2018; Erickson et al. 2018).

The terms we use to define those who do not migrate can obscure the idea that this status can be the result of a positive or deliberative choice. Mobility and migration communicate the idea of action; immobility and non-migrant do the opposite. Left-behind is perhaps the most negative of terms to use to describe those who do not move to places with better economic opportunities (Eisenberg, 2024). Stayer is somewhat free of these associations, suggesting a degree of agency (Schewel 2019). Long-term or lifetime stayers are possible descriptors of the subjects of our investigation, but we opt to use "never migrants" and "never migrating" to describe them and their behavior, using immobile and immobility as substitutes. These terms best capture what we mean by the subjects of our study even though we recognize they may, for some, signal that remaining in place is defined in relation to its opposite, migration or mobility. We consider never migrating a positive or deliberate choice for some.

U.S. immobility at the interstate scale: conceptualizations and research questions

We conduct our empirical investigation of never migrating at the state scale, considering someone who never migrates to have lived in the same state for all the years we observe them in the data, 2001-2016. The choice of states is partly motivated by data disclosure procedures for the restricted-use administrative data we use. When migration units have populations smaller than the least populous congressional district (Wyoming's state-wide at-large district as of this writing), there is a greater barrier to disclosure of results, possibly necessitating noise injection. Regardless, states are an important scale for assessing U.S. internal migration, with the rate of migration between them part of a commonly reported suite of measures at different geographic scales.

Migration rates depend on the geographic scale of units in which they are measured (Bell *et al.*, 2015b). The smaller the area of the unit, the higher the rate because most people migrate short distances (Long *et al.*, 1988; Molloy *et al.*, 2011). As Barcus and Brunn (2010) note, the scale at which we choose to define immobility is subject to the same effect. They argue for a regional approach to the study of Appalachian place attachment, framing immobility as staying within a culturally defined region rather than as the absence of any move. A strict definition would limit immobility to never undertaking any migration, excluding from the category of immobile even

those who undertake short-distance moves within a local area (Hjälm 2014). Zelinsky (1983) notes that nobody would fall within the category of never migrant if migration is defined as any movement at any scale. What matters, he suggests, is whether a move ruptures the mover's social relationships.

Schewel's (2019) definition of immobility offers a way forward. Building on Hägerstrand's (1957) idea that migration involves a shift in one's center of gravity, Schewel defines immobility as spatial continuity in that center of gravity. This center could be a single residence from which people go about their everyday life. A more expansive notion of center could allow for permanent residential moves within an area that is a person's daily activity space (Thomassen *et al.*, 2024). Such moves do not disrupt regular local connections to family and friends or to labor market and cultural resources (Roseman 1971). As migration distance increases these connections may not be daily but still regular enough to be sustained, nurturing a sense of belonging within a larger cultural region. This aligns with the approach of Barcus and Brunn (2010) and Zelinsky's (1983) observations that longer-distance migrations do not necessarily rupture social connections. We approach our study of state-scale immobility from this expansive meaning of center of gravity.

Our investigation focuses on those who never migrate out of state between 2001 and 2016 for a cohort between the ages of 15 and 50 in 2001 (30 and 65 in 2016). This cohort includes people who transition into young adulthood and those who transition from their middle working years to retirement age during the 16 years we follow them. Our aim is to provide what we believe to be the first measurement of national immobility at the state scale for these key segments of the population.

We have access to information on age, sex, race, and nativity (U.S.-born vs. foreign-born). Our expectation is that the rate of immobility will be lowest for the youngest in our group who pass through the peak years of migration in the 16 years we follow them and will increase with age following an inverse pattern to the typical migration schedule (Rogers *et al.*, 1978). It is difficult to anticipate variation in immobility rates by sex, race and nativity. These groups do vary in their migration rates: marginally lower for females than males, lower for Black and Hispanic people than White people, and lower for the foreign-born than U.S.-born (Molloy *et al.*, 2011). The most likely outcome is that immobility rates replicate this pattern because groups that are most likely to move annually are *probably* least likely to never migrate.

We have no information on education, but because our data are from tax returns they do record

income that is not top-coded or censored. This permits flexibility in calculating immobility rates by income and we opt to do so by decile as measured in tax year 2000, i.e., prior to the period of observed immobility. A key advantage of the tax data is that we can find this income for parents or guardians of those reported as child dependents on tax returns. This allows us to associate income deciles with children who do not have a tax return of their own in 2000 because they have yet to enter the labor force (e.g., 15-year-olds). We can then assess how family income affects the likelihood of a teenager never migrating from the state they lived in with their parents as a teenage dependent in 2001 through to their early thirties in 2016.

As with the sociodemographic variables, the expected effect of income on never migrating across state lines is unclear. There is evidence that poor families are apt to move more frequently than higher income groups because they are more likely to experience residential instability (Desmond *et al.*, 2015; DeLuca *et al.*, 2019). However, it is not clear if this instability spills into moves across state lines, yielding a low immobility rate for the poor relative to other groups at the state scale. Self-reported income data suggest interstate migration rates are marginally lower for those in the bottom than top half of the income distribution (Molloy *et al.*, 2011). Thus, a priori, it seems possible that the bottom decile(s) might have higher rates of never migrating than those in top deciles. We now turn to a more detailed description of our data sources and the procedures we used to produce a linked dataset and select records for empirical analysis.

DATA

The project uses annual administrative tax record microdata from the Internal Revenue Service (IRS) that, when linked over time, allow for a longitudinal analysis of migration that is novel in terms of detail and scale for the U.S.. As such, these data provide opportunities for more comprehensive assessments of migration in the U.S. than existing data sources (Foster *et al.*, 2018a). The use of administrative records for migration research in the U.S. parallels developments in other countries (Ernsten *et al.*, 2018; Foley *et al.*, 2021; Bernard *et al.*, 2024). These U.S. and other country studies show that administrative data have much potential for migration and other socio-demographic research but may have selectivity issues and may not include sufficient socio-demographic variables for in-depth analyses. The data we use include additional tax information to minimizes selectivity issues and we link them to other administrative and survey data sources to provide socio-demographic information for each person (Foster *et al.*, 2018a).

Our project specifically uses two sources of Federal Tax Information (FTI) from the IRS, one source from the Social Security Administration (SSA), and one source from the Census Bureau.[1] The first source of FTI, IRS 1040s for the tax years 2000 through 2015, is our primary source of longitudinal information. These records are scrubbed of Personally Identifiable but retain much of the data individuals report when filing income taxes each year including geographic indicators and incomes.

The second source of FTI, IRS 1099 "information returns" for tax years 2003 to 2015 that are created by the IRS to flag sources of income for individuals. They provide insight into a population typically overlooked in studies of migration relying solely on 1040s – namely, those who do not and/or are not required to file income taxes. Publicly available IRS aggregate migration flow data only use locational information from 1040s, excluding those who do not file. The addition of 1099 information returns mitigate these exclusions, reducing selection issues associated with the public data (Foster, Ellis and Fiorio 2018a, 2018b).

All PII such as names, Social Security Numbers (SSNs), and Individual Taxpayer Identification Numbers (ITINs) are removed from administrative records to protect privacy and reduce the risk of disclosing personal information. They are replaced with unique, anonymous personal identifiers which facilitate individual-level record linkage across data sources. These unique identifiers are assigned on SSNs or ITINs, when available, and probabilistically assigned using other PII, such as date of birth and address, when an SSN or ITIN is not available (Wagner and Layne 2014). Over 99 percent of individuals found in IRS administrative records receive a unique person identifier because SSNs or ITINs are required on tax documents.

Given that our primary focus is on interstate mobility, we use PIKs to link individual-level state information from the 1040 and information returns longitudinally across tax years. We give preference to 1040 state information. If an individual appears in the 1040 data for a particular year, as a filer or as a dependent, we take the state associated with the filing and attribute it to the individual for that year. If the individual is missing a 1040 for a particular year, we use state information from the information returns (1098/1099) files where it is available. Because an individual can receive multiple information returns associated with different states in a given

[1] Access to these sensitive administrative records sources was granted by the Internal Revenue Service and U.S. Census Bureau under the authority of Title 13 and Title 26 of U.S. Code and following research approval at the Center for Economic Studies at the Census Bureau. Given confidentiality concerns, data used in this analysis are not available for public use.

year, we determine the most commonly occurring state for each individual year.

The result is file with sixteen possible state observations from 2001 to 2016 for each individual from the 2000 to 2015 tax year files. The state observations run a year ahead of the tax years because they record where people were when they filed taxes in the year after the tax year ends. Changes in tax-based state location in the filing year are the basis for our assessment of immobility. It is worth noting that changes in the state where someone files taxes or receives a 1099 will likely coincide with changes in their place of usual residence, the location-standard used for determining migration in surveys, such as the American Community Survey (ACS). For some, however, tax-based measures of location change will not coincide with survey-based measures of changes in place of usual residence. This does not mean one measure of migration change is better; it means they are different measures of migration.

To better understand the relationships between interstate immobility and family income we draw on adjusted gross income from the 1040 data. For any given tax year, we can link individuals, including filers, spouses, and dependents, to the adjusted income associated with the file in which they appear. We call this family income.

We link tax data to data from SSA in the Numerical Identification Database (Numident), a cumulative dataset containing all individuals ever assigned an SSN. The Numident provides information on date of birth and death, place of birth including country for those born outside the U.S., citizenship status, and sex. The resulting dataset only includes records from tax filers with SSNs because the Numident supplies their demographic information, which allows us to distinguish migration sequences among population subgroups and to identify censoring due to mortality. Those who file with Individual Taxpayer Identification Numbers (ITINs) are not in our sample because we have limited demographic information on them.

Finally, we obtain race and Hispanic information for filers and their dependents by linking these data to the Census Bureau's Best Race and Ethnicity Administrative Records file, which uses a set of rules to assign "best" race and "best" Hispanic responses from a composite of data sources, including the decennial census and Numident but also a number of other federal, state, and third-party survey and administrative sources (Ennis *et al.*, 2018; Noon 2024).

We categorize someone as never migrating or immobile if they are recorded as in the same state in all the years we observe them. This is straightforward when there is a record for them every year, which describes 64.1 percent of our records, but there is a long tail of people with

incomplete records. Figure 1 shows the percentage of individuals missing one or more tax year, excluding those who were born after 2000 or who had died before 2016 (the year 2015 taxes were filed).

As Figure 2 demonstrates, there is a prominent non-linear relationship between number of years missing and the probability of being categorized as never migrating. It suggests we face a trade-off. Including only individuals with complete tax information (i.e. 16 years of reporting) would likely overestimate stayers with respect to movers. This makes sense if one considers that people who do not move are much more likely to have stability in their home and work life, conditions that favor filing a tax return. Including those with only one year of data (1.7 percent of records) is meaningless because they can only be immobile. Moreover, given the slightly reduced coverage of the information returns (tax years 2003 to 2015), individuals who only appear in the information returns are missing exactly three years (hence the anomalous trend line), and it would be self-defeating to exclude them.

For these reasons, we aim for the middle ground, choosing to include anyone with ten or more tax years (six or less years missing) over the 16 years of data. This allows us to include 89.6 percent of records in which immobility is not pre-determined (i.e., excluding those observed in only one year). It also means we work with data in which everybody is observed in sequential years all or most of the time, with minimal gaps for a minority of records and thus a low risk of wrongly inferring immobility because of an unrecorded move. After applying these missing year conditions for those between 15 and 50 in 2001 who were alive in 2016, we end up with 135,100, 000 unique records (rounded for disclosure) in our sample cohort for analysis.

ANALYSIS

The presentation of the analysis is in three stages. We begin by reviewing immobility rates for our study population and by sociodemographic groups and income. The subsequent stage reports on a multivariate model that measures the relative impact of each sociodemographic characteristic and income on the probability of being immobile. The model estimates measure the effect of each characteristics on the immobility rate controlling for group differences in the distribution of other characteristics. The final stage reports on the geography of state immobility rates.

Descriptive findings

Table 1 shows the relative share of all people between 15 and 50 years old in 2001 who never

migrated across state lines between 2001 and 2016. For the total sample as a whole, 75 percent never migrated between states from 2001 and 2016. To put this in perspective, we can compare it to an immobility rate estimated from annual interstate migration rates from the same data source. These data yield an average interstate migration rate of 2.94 percent for tax years 2008-2012 (Foster *et al.*, 2018). From this rate we can crudely estimate the proportion who never migrate over the 16 years in our study (15 years of annual migration): $(1\text{-}0.0294)^{15} = 0.6204$, or 62.04%. This estimate assumes everyone has the same probability of moving each year, which research has repeatedly shown to be false: those who have migrated before are more likely to move again (DaVanzo and Morrison 1981). The result is that annual migration rates yield a considerable underestimate of the rate of never migrating because of heterogeneity within the population in the probability of migrating.

There are variations in these rates of never migrating by race, nativity and sex. These are generally small but there are some noteworthy differences. Females have a higher immobility rate than males, which is consistent with them migrating between states at a lower rate than males (Molloy *et al.*, 2011). Among the four largest race and ethnicity categories, Hispanics are the most immobile, followed by Whites, Blacks, and Asian/Pacific Islanders (Asian/PI) in decreasing rates of immobility. This ranking partially matches what we anticipate from racial differences in interstate migration rates (Molloy *et al.*, 2011). Hispanics have the highest rate of never migrating, which accords with them having the lowest interstate migration rate among major groups. The Black rate of never migrating is lower than for Whites but this is not what we would anticipate from the Black interstate migration rate being lower than for Whites (Molloy *et al.*, 2011). This apparent inconsistency is possibly the result of differences between these two groups in the degree of heterogeneity within their populations in the probability of migration, with a sharper distinction between mobile and immobile subgroups in the White population. Specifically, Whites could have a higher migration rate and a higher rate of never migrating than Blacks if they also have a higher rate of repeat migration than Blacks. The distinctive geographies and migration trends of these two groups could also be a factor, with Black Great Migration populations and their descendants who have lived in northern states for much of their lives moving south (Hunt *et.al.,* 2008, 2012).

Switching to nativity, the immobility rate for the foreign-born is higher than for the U.S.-born. Because our data only capture immobility for the foreign-born with SSNs (i.e. it does not include Individual Tax Identification Number (ITIN) filers) there is a need for caution in interpreting this result. Nevertheless, the finding is consistent with the foreign-born having a lower migration rate than the U.S.-born (Molloy *et al., 2011)*. The secondary migration of

immigrants within the U.S. has garnered considerable attention as a component of the growth in foreign-born populations in new destinations (Lichter and Johnson, 2009). This dispersion slowed as the 2007-9 financial crisis unfolded and jobs in new destinations in traditional immigrant employment sectors, such as construction, disappeared (Ellis *et al.* 2014). Thus, during the period we observe the population, the tendency for the immigrant population to stay where they are may have increased.

Figure 3 shows the fifteen-year immobility schedule by age in 2001. As expected, this plot looks like a reflected and slightly transposed version of the well-established annual mobility schedule by age (Rogers *et.al.* 1978). Individuals who were 19 in 2001 (34 in 2016) were the least immobile over the period 2001 to 2016 and immobility increases nonlinearly with age up to 45 in 2001 (60 in 2016). It then drops slightly, probably because of early retirement migration by those who were in their late 40s in 2001 who age into their early 60s by 2016. The immobility rate even for the least immobile is high, over 62% for 19-year-olds, meaning that even those who transition through their most mobile years are more likely to never migrate from the state they were in as a teenager than migrate.[2]

Income increases with age and thus to explore the effect of income on immobility uncontaminated by the age effects in figure 3, we chart the 2001-2016 never migrating percentage for those who were in four one-year age categories in 2001 (15, 25, 35, 45) by family income decile in 2000 (figure 4). As one would expect, these curves affirm that immobility is highest for those in older ages. The peak level of immobility for the three oldest cohorts occurs in the eighth decile; for the youngest cohort it occurs at the fourth decile. At all ages, the top (10th) and bottom (1st) deciles have lower rates of never migrating than do deciles in the middle of the income distribution. Thus, immobility follows an inverted U-shaped distribution by income: it is not the experience of the poorest and richest; it is of the middle or upper middle-income classes, though there are differences in the configuration of this experience at the youngest age. We cannot ascertain why the poorest and richest deciles have lower immobility. The reasons likely differ for each, but our findings suggests that neither group is best

[2] Late teenage immobility is possibly overestimated because a subset of those who move to another state for college may not be recorded in the tax data as interstate migrants. Out of state college students are a minority of all teenagers and it is only the fraction of this group that never files their own 1040 from their college state, is claimed as a dependent in their origin state, and returns to their origin state permanently after graduation who we will erroneously categorize as having never migrated. Thus, the overestimation of late teenage immobility is probably small.

characterized as either stuck or rooted.

For the two older age cohorts, those who transition toward retirement or into later working life during the period we observe them, the least immobile are those at the bottom of the income distribution; they are eight percentage points less likely never migrate than those in the eighth decile, the most immobile decile for both of these cohorts. Those in the top decile are five percentage points less immobile than the eighth decile for 35 and 45-year-olds.

The two younger cohorts differ from the older cohorts in that they are least immobile in the top not the bottom decile. The younger cohorts also have larger peak-to-trough immobility differences across the income distribution. Twenty-five-year-olds in the tenth decile are 14 percentage points less likely to be immobile than peers in the eighth decile, the least immobile income decile for this age cohort; 15-year-olds in the top decile are 12 percentage points less immobile than those in their most immobile decile, the fourth. The equivalent drops to the bottom decile are smaller: nine percentage points for 25-year-olds and eight percentage points for 15-year-olds.

Some of the young vs. old cohort differences could originate in parental vs own income effects, at least for the 15-year-old cohort. For those who were 15 in 2001, we are measuring the effect of parental income on their 16-year immobility rate whereas for others we are capturing the effect of their own income. Thus for 15-year-olds we have a possible intergenerational effect that elevates the immobility rate of those at lower parental incomes and reduces it at higher incomes. Whatever the cause, the variation in immobility rate across deciles within each age cohort is greater than the differences in immobility by race, sex, and nativity. Economic circumstances matter more for immobility than socio-demographic characteristics.

That the decile immobility rate profile for 25-year-olds generally overlaps with that for 15-year-olds (except for the different decade of peak immobility) suggests a possible period rather than age-specific effect for these two groups. These two younger cohorts experienced the dislocations of the 2007-9 financial crisis and its aftermath just after they entered the labor market or when they were in their early thirties. The employment instability this generated at the beginning of their careers may have generated different interrelated mobility and earnings effects on them than on older working adults with more secure jobs.

Multivariate findings

We next explore these results in a multivariate model to assess whether they are affected by controls. We estimate a linear probability model that predicts whether someone never migrates as a function of their sociodemographic characteristics and income decile. We estimate two versions of this model, with and without state fixed effects, to assess whether the estimates are sensitive to state differences in immobility rates (we discuss the geography of state immobility rates in the next section). The linear probability model's coefficients provide direct estimates of the marginal effects of each variable on the outcome of interest: the probability of never migrating. Standard errors are corrected for heteroscedasticity, though this has little effect on precision because of the large N.

Table 2 reports the model estimates. As with the descriptives, the model is estimated on those aged 15 to 50 in 2001 but restricted to those for whom we have an observed family income in 2000 to ensure estimation of decile effects. As before, age is measured in 2001 but transformed by subtracting 15. Age squared is included to account for the non-linearity observed in Figure 3. The age transformation and the selection of reference categories (males, White, U.S.-born, first income decile) means the model's intercept is an estimate of the probability of never migrating for U.S.-born White males who were 15 years old in 2001 and at that time were in a family with income in the first decile. Model 1's estimates are without state fixed effects; model 2 includes them. The inclusion of state fixed effects has very small effects on most reported coefficients, with the largest changes in the effects of race. In general, the magnitude and relative size of the effects remain almost the same across the models.

The model coefficients align with the descriptives for some effects but not others. The coefficients suggest immobility non-linearly increases with age and that females are more immobile than males, which replicates the descriptive findings. The income coefficients, which suggest the bottom (omitted) and the top deciles have the lowest rates of immobility and highest immobility rate is for the 6th decile, is similar to the inverted U-shaped distributions in Figure 4.

The nativity and race group coefficient differences do not line-up with the descriptive findings, however, suggesting they are sensitive to controls for other characteristics. The model coefficients suggest the foreign-born are 1.4 to 1.7 percent less immobile than the U.S.-born whereas the descriptives reported them as 1.7 percent more immobile. The descriptive immobility rates for Asian and Black people are lower than those for White people (the omitted

category) whereas the model estimates indicate they are higher than for Whites, though only marginally so in the case of Asians. This means that the model estimates have Whites with the lowest immobility rate, not Asians. Hispanics are the most immobile group in the model estimates just as they are in the descriptives. As such, they are the only group whose relative immobility rate ranking is stable across models and descriptives. It is worth noting that the ranking of racial group differences in immobility from the model estimates is more consistent with the ranking of racial group differences in interstate migration rates than from the descriptives.

State differences in immobility rates

Figure 5 shows that state immobility rates vary considerably and have a distinctive geographical pattern. In Alaska, Nevada, and Wyoming, these rates are below 65 percent. In Texas, Wisconsin and Pennsylvania they are above 80 percent. Western states have below average levels of immobility, eastern states have above average levels of immobility.

There are key exceptions to this regionalization. California has well-above-average immobility rates and stands apart from its Western neighbors. For all the talk of a California exodus, its population has an above-average share of residents in 2001 who stayed there through 2016. In this it is more like Texas than neighboring western region states. The low immobility rate exceptions in the east are likely artifacts of small states with boundaries bisecting metropolitan labor markets generating substantial residential mobility across state borders. Rhode Island and southern New Hampshire are within the Boston region's labor market. Parts of Maryland and Virginia are within the greater Washington and Baltimore labor market.

The map identifies regions with known place attachment, such as the Appalachian states of Kentucky, West Virginia and their neighbors, that exhibit a relatively high rate of never migrating. Appalachia is not unusually immobile, however, as these rates are matched or exceeded in the upper midwestern states of Michigan, Wisconsin and Minnesota. The high rates of immobility in Texas and California may be partly due to having large shares of Hispanics with high immobility rates. However, a large Hispanic share does not guarantee a low rate of never migrating, e.g., New Mexico. Southern states that have attracted large numbers of migrants, such as Georgia and Florida, have lower rates of never migrating than their neighbors, possibly because their large migration inflows have yielded resident populations with reduced attachment to the state. However, other southern states with large inflows (e.g., North Carolina) have higher immobility rates than Georgia. North Carolina's rate of never migrating

is, in fact, similar to that in some midwestern states with no recent history of large inflows. These differences and others on the map suggest the state geography of immobility is not reducible to demographic compositional explanations.

CONCLUSIONS

In this paper, we used a U.S. population-sized longitudinal dataset built from IRS and Social Security administrative records to investigate interstate immobility. These data show that over 75 percent of people who were between 15 and 50 years old in 2001 never migrated across state lines by 2016. As expected, this percentage varies by age. Yet, even for those who were teenagers in 2001 and pass through the most migratory period of their lives during the subsequent 16-year period we observe them, over 62 percent never left their state. We have no information to assess whether this level of immobility has changed over time, but our information does mean that never migrating across state lines for the 16-year period of our analysis is the prevailing experience of the U.S. population. At this scale at least, most Americans stay put, even the young adults who we least expect to do so.

Nativity and sex have modest effects on the immobility rate, under 1.7 percentage points with the precise amount depending on how it is measured. The differences between racial groups are more substantial with a gap between the most and least immobile major groups of about six percentage points, again depending on how it is measured. The Hispanic population is the most immobile group and the Asian/PI population the least regardless of univariate or multivariate measurement approach.

One of the strengths of our data is information on income and it, measured as family income in 2000 at the beginning of the analysis period, has a marked effect on the rate of immobility. The peak rate of immobility for cohorts aged 25 and above is at the eighth income decile with the lowest rates at both extremes of the income distribution. The same inverted U-shaped distribution exists for the youngest cohort, but their peak immobility rate is at a lower decile. Regardless of age, the rich and poor are the economic classes least likely to be rooted or stuck. Whether by choice or not, and it seems most likely this is a choice that the rich can exercise more than the poor, these two economic extremes appear to be the most detached from place. Among the adult cohorts, the most attached to their state are the prosperous middle classes, if that is what their relatively high rate of immobility represents. Our data cannot tell us why there are these income differentials in immobility, but they do challenge narratives of immobility that center on the inability or unwillingness of less well-off populations to move.

Those who are the least well-off are more apt to fit the caricature of restless Americans than the prosperous middle classes.

Our data cannot tell us whether the rate of never migrating has changed from previous decades. It is possible that immobility is more prevalent now than a few decades ago when interstate migration rates were twice their current levels. The nature of immobility measurement, requiring observations over lengthy periods of time to distinguish those who are currently immobile, means that observation of trends in immobility will require more decades of data than currently available.

While we cannot observe variations in the rate of never migrating across time we can do so across spatial scale. The state scale of our immobility analysis mirrors that used in many studies of U.S. migration. Future research can use these data to compare levels of immobility at finer scales, such as commuter zone, county and zip code. These measurements will be useful for gauging the relationship between immobility and scale and identifying local geographies of immobile populations.

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	Share of Sample	Percent Ever Migrates	Percent Never Migrates
Total	100	25	75
Sex			
Male	49.2	25.8	74.2
Female	50.8	24.3	75.7
Race			
White	66.6	24.9	75.1
Black	11.7	26.3	73.7
Asian/PI	4.8	27.4	72.6
Nat. Am.	0.9	28.3	71.7
Other	0.2	25.7	74.3
Hispanic	13.4	21.6	78.4
Mixed	1.2	31.9	68.1
Missing	1.2	36.4	63.6
Nativity			
U.S.-born	84.9	25.2	74.8
Foreign-Born	15.1	24.1	75.9

Table 1: Sample shares and percent ever/never migrating between 2001 and 2016 by sex, race and nativity for those between 15 and 50 years of age in 2001.

Data sources: IRS 1040s (2000-2015), IRS 1099s (2003-2015), Numident, Census Best Race and Ethnicity Administrative Records. N= 135,100, 000 unique records (rounded for disclosure). Disclosure Review Board release approval: CBDRB-FY24-CES005-016.

	No state fixed effects			State fixed effects		
	Coef	**SE**	**CI**	**Coef**	**SE**	**CI**
Constant	0.5331	0.0002	(0.5327, 0.5333)	0.5379	0.0002	(0.5375, 0.5383)
Age	0.0112	0	(0.0112, 0.0112)	0.0113	0	(0.0113, 0.0113)
Age²	-0.0001	0	(-0.0001, -0.0001)	-0.0001	0	(-0.0001, -0.0001)
Female	0.0104	0.0001	(0.0102, 0.0104)	0.0105	0.0001	(0.0103, 0.0107)
Foreign-born	-0.0169	0.0001	(-0.0171, -0.0169)	-0.0141	0.0001	(-0.0143, -0.0139)
Race (White ref)						
Black	0.0073	0.0001	(0.0071, 0.0073)	0.0118	0.0001	(0.0116, 0.012)
Asian/PI	0.0090	0.0002	(0.0086, 0.009)	0.0035	0.0002	(0.0031, 0.0039)
Nat. Am.	-0.0116	0.0004	(-0.0124, -0.0116)	0.0132	0.0004	(0.0124, 0.014)
Other	-0.0159	0.0008	(-0.0175, -0.0159)	-0.0126	0.0008	(-0.0142, -0.011)
Hispanic	0.0630	0.0001	(0.0628, 0.063)	0.0556	0.0001	(0.0554, 0.0558)
Mixed	-0.0328	0.0004	(-0.0336, -0.0328)	-0.0284	0.0004	(-0.0292, -0.0276)
Missing	-0.0758	0.0009	(-0.0776, -0.0759)	-0.0757	0.0009	(-0.0775, -0.0739)
Family income decile (1st ref)						
2nd	0.0495	0.0002	(0.0491, 0.0495)	0.0486	0.0002	(0.0482, 0.049)
3rd	0.0559	0.0002	(0.0555, 0.0559)	0.0552	0.0002	(0.0548, 0.0556)
4th	0.0599	0.0002	(0.0595, 0.0599)	0.0593	0.0002	(0.0589, 0.0597)
5th	0.0692	0.0002	(0.0688, 0.0692)	0.0684	0.0002	(0.068, 0.0688)
6th	0.0759	0.0002	(0.0755, 0.0759)	0.0748	0.0002	(0.0744, 0.0752)
7th	0.0722	0.0002	(0.0718, 0.0722)	0.0711	0.0002	(0.0707, 0.0715)
8th	0.0498	0.0002	(0.0494, 0.0498)	0.0491	0.0002	(0.0487, 0.0495)
9th	0.0350	0.0002	(0.0346, 0.035)	0.0336	0.0002	(0.0332, 0.034)
10th	0.0127	0.0002	(0.0123, 0.0127)	0.0113	0.0002	(0.0109, 0.0117)

Table 2: Model predicting never migrating between 2001 and 2016 for those between 15 and 50 years of age in 2001.

Notes: Linear probability model estimates predicting the probability of never migrating, 2001-2012 for those between 15 and 50 years old in 2001. Standard errors corrected for heteroscedasticity. State fixed effects not reported. *Data sources:* IRS 1040s, IRS 1099s, Numident., Census Best Race and Ethnicity Administrative Records. N= 118,300,000 unique records (rounded for disclosure). Disclosure Review Board release approval: CBDRB-FY24-CES005-016.

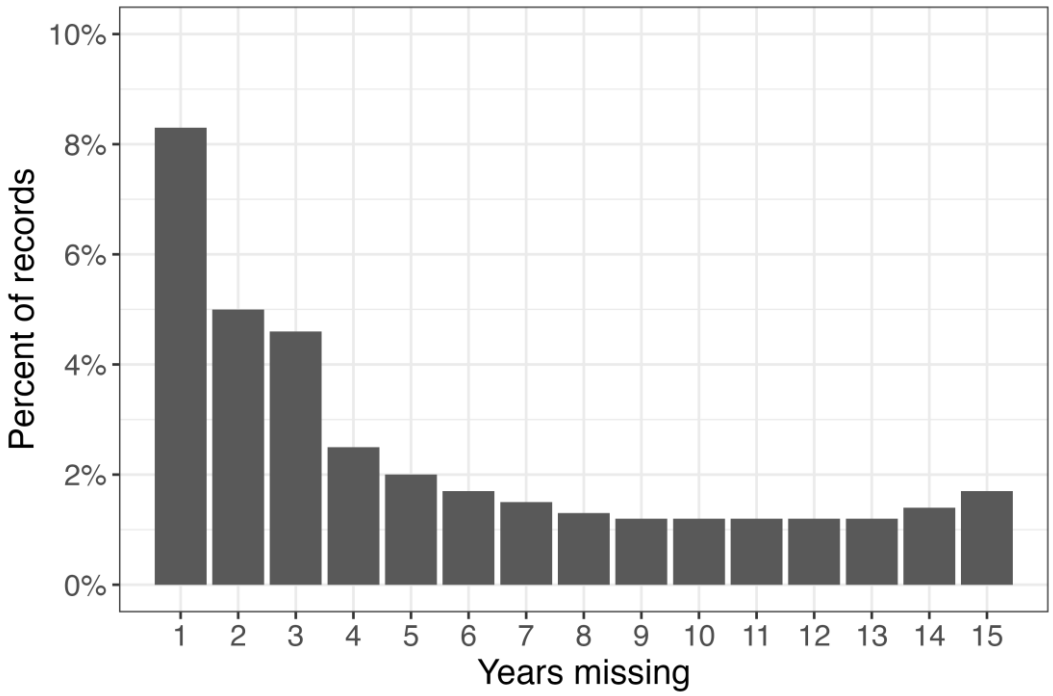


Figure 1. Percent of records by number of years missing (>0 and <16).

Notes: Each bar records the percent of records missing a number of tax years e.g. 8.3% of records are missing one tax year. *Data sources:* Numident, IRS 1040s (2000-2015) and 1099s (2003-2015), N=153,400,000 unique records (rounded for disclosure). Disclosure Review Board release approval: CBDRB-FY24-CES005-016.

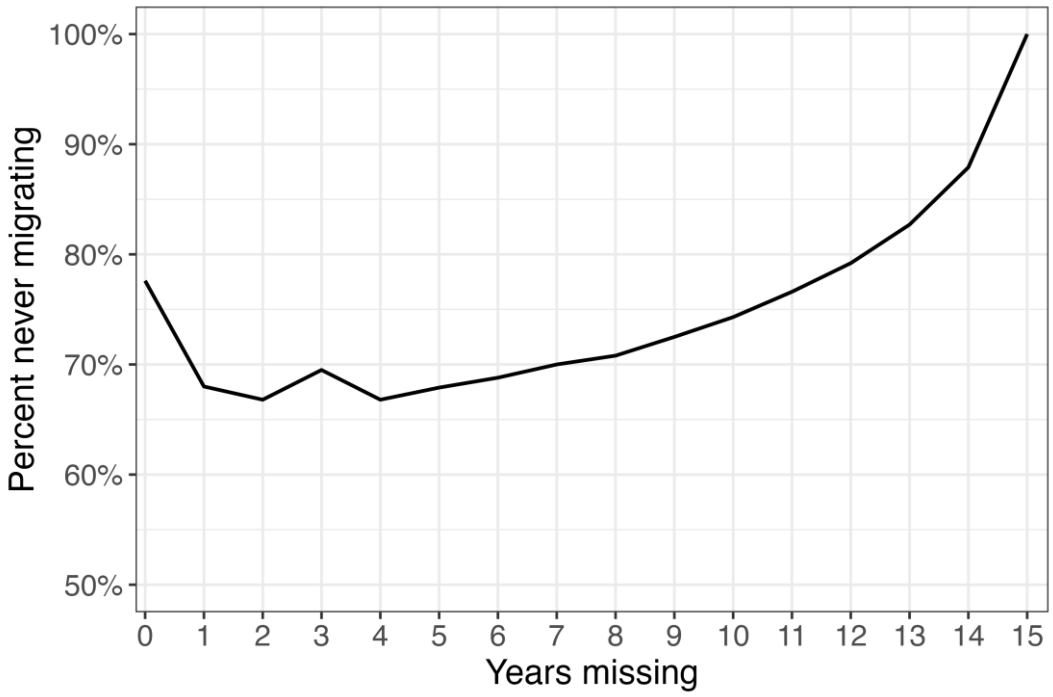


Figure 2. Percent never migrating 2001 to 2016 by number of years missing in the data

Notes: This chart shows the percentage never migrating between 2001 and 2016 by years missing. For example, for records with 0 years missing (i.e. 16 years of data) the percentage never migrating is 77.6 percent. The anomaly at three years missing arises from the three-year difference in number of years of coverage between 1040s (2000-2015) and 1099s (2003-2015). *Data sources:* Numident, IRS 1040s (2000-2015) and 1099s (2003-2015), N=153,400,000 unique records (rounded for disclosure). Disclosure Review Board release approval: CBDRB-FY24-CES005-016.

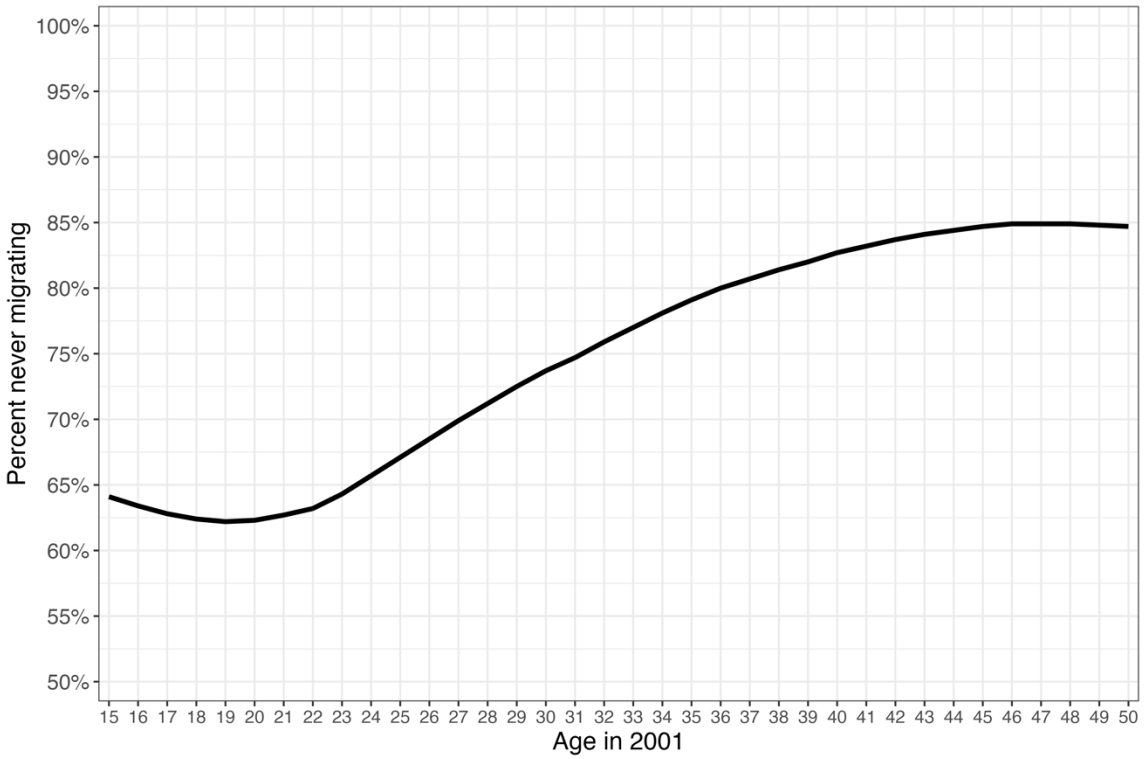


Figure 3: Percent never migrating 2001 to 2016 by age for those between 15 and 50 years of age in 2001.

Notes: This chart shows the percent never migrating between 2001 and 2016 by age as recorded in 2001. *Data sources:* Numident, IRS 1040s (2000-2015) and 1099s (2003-2015), N=135,100,000 unique records (rounded for disclosure). Disclosure Review Board release approval: CBDRB-FY24-CES005-016.

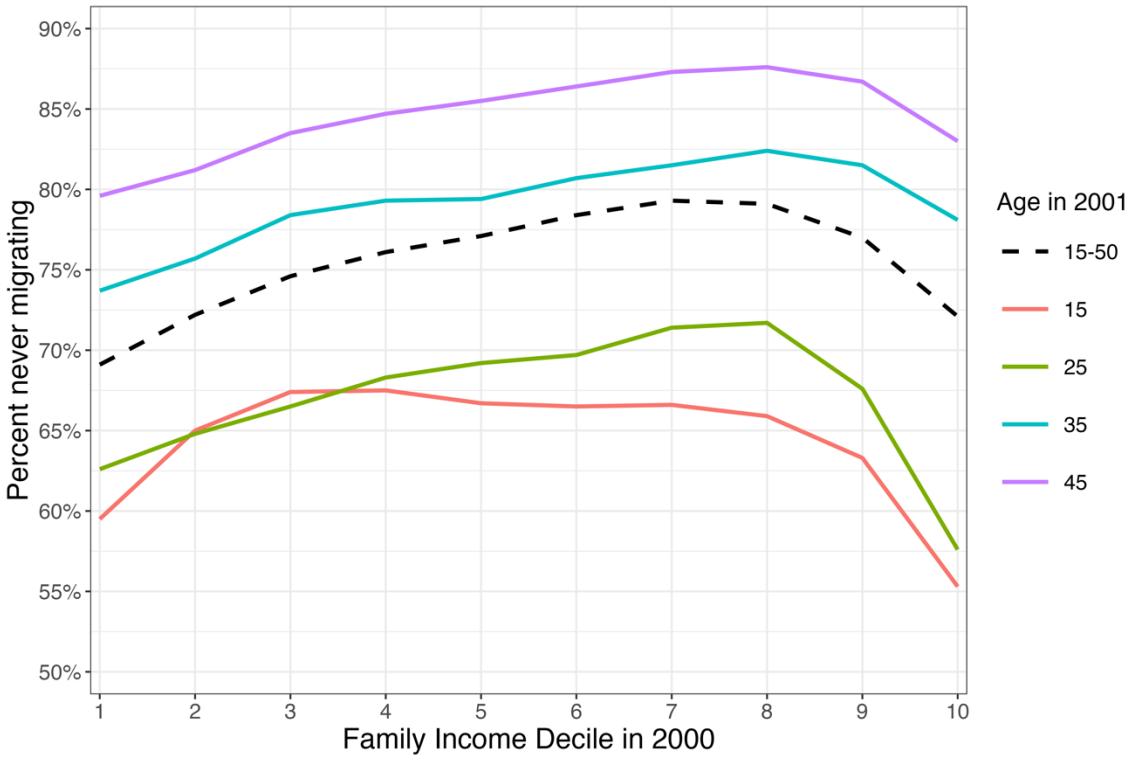


Figure 4: Percent never migrating 2001-2016 by family income decile by age in 2001.

Notes: This chart shows the percent never migrating by family income decile in 2000 for the complete sample (i.e., those aged 15-50 in 2001) and four age cohorts (those aged 15, 25, 35, and 45 in 2001). *Data sources:* Numident, IRS 1040s (2000-2015) and 1099s (2003-2015), N=118,300,000 unique records (rounded for disclosure). Disclosure Review Board release approval: CBDRB-FY24-CES005-016.

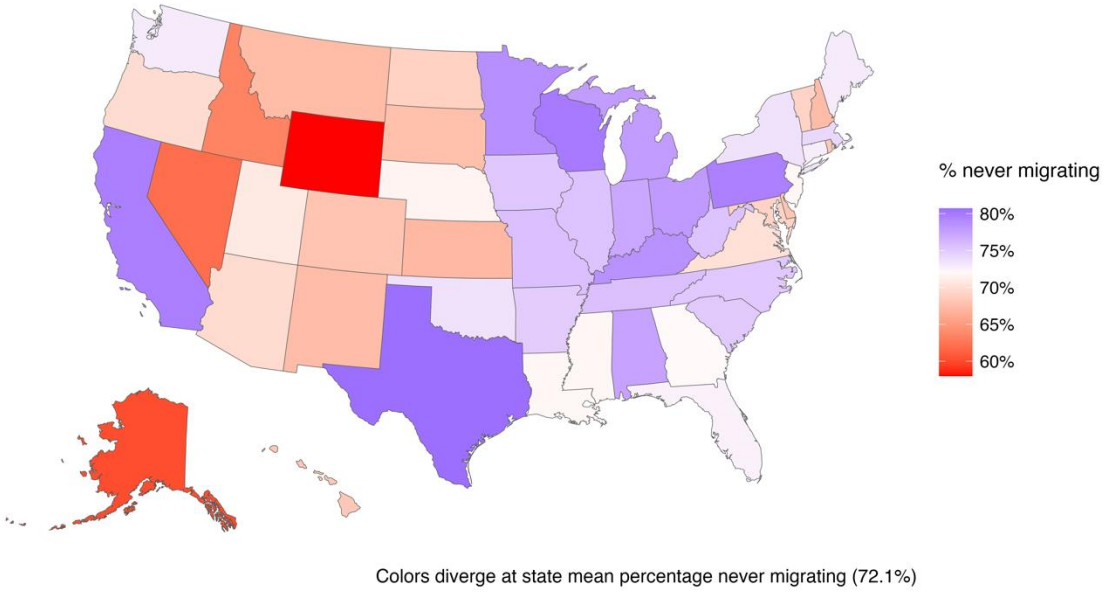


Figure 5: Percent never migrating between 2001-2016 by state for those between 15 and 50 years of age in 2001.

Data source: Numident, IRS 1040s (2000-2015) and 1099s (2003-2015), N=135,100,000 unique records (rounded for disclosure). Disclosure Review Board release approval: CBDRB-FY24-CES005-016.